In [1]:

```
import numpy as np
import pandas as pd

# 1. Create a DataFrame from a list
data = [
 [1, 'Smith', 50000],
 [2, 'Jones', 60000]
]

df = pd.DataFrame(data)
df.columns = ['Empd', 'Name', 'Salary']

print("Employee DataFrame:")
print(df)

# 2. Display DataFrame information
print("\nDataFrame Information:")
df.info()

# 3. Create an employee dataset
employee_data = {
 'emp id': [1, 2, 3, 4, 5, 6, 7],
 'name': [
 'SREE VARSSINI K S',
 'SREEMATHI B',
 'SREYA G',
 'SREYASKARI MULLAPUDI',
 'SRI AKASH U G',
 'SRI HARSHAVARDHANAN R',
 'SRI HARSHAVARDHANAN R'
 ],
 'salary': [5000, 6000, 7000, 5000, 8000, 3000, 6000] }

df = pd.DataFrame(employee_data)

# 4. Display the first and last five rows
print("\nFirst Five Rows:")
print(df.head())

print("\nLast Five Rows:")
print(df.tail())

# 5. Display DataFrame information
print("\nEmployee DataFrame Information:")
df.info()

# 6. Select the salary column
print("\nSalary Column:")
print(df['salary'])

# 7. Calculate statistical measures
print("\nMean Salary:", df['salary'].mean())
print("Median Salary:", df['salary'].median())
print("Mode Salary:", df['salary'].mode().tolist())
print("Salary Variance:", df['salary'].var())
print("Salary Standard Deviation:", df['salary'].std())

# 8. Display numerical summary
print("\nStatistical Summary:")
print(df.describe())

# 9. Display summary of all columns
print("\nComplete Summary:")
print(df.describe(include='all'))

# 10. Display column names
print("\nColumn Names:")
print(df.columns)

# 11. Convert DataFrame into a NumPy array
emp_array = df.values
print("\nEmployee Array:")
print(emp_array)

# 12. Create a DataFrame from the NumPy array employee_DF
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```
 = pd.DataFrame(emp_array, columns=df.columns)

 print("\nFinal Employee DataFrame:")
 print(employee_DF)
```

```
Employee DataFrame:
 Empd Name Salary
0 1 Smith 50000
1 2 Jones 60000

DataFrame Information:
<class 'pandas.DataFrame'>
RangeIndex: 2 entries, 0 to 1
Data columns (total 3 columns): #
Column Non-Null Count Dtype ---
------ -------------- ----- 0 Empd
2 non-null int64 1 Name 2 non-null
str  2 Salary 2 non-null int64
dtypes: int64(2), str(1)
memory usage: 190.0 bytes

First Five Rows:
 emp id name salary 0 1 SREE VARSSINI K
S 5000 1 2 SREEMATHI B 6000 2 3 SREYA G
7000 3 4 SREYASKARI MULLAPUDI 5000 4 5
SRI AKASH U G 8000

Last Five Rows:
 emp id name salary 2 3 SREYA G 7000 3 4
SREYASKARI MULLAPUDI 5000 4 5 SRI AKASH
U G 8000 5 6 SRI HARSHAVARDHANAN R 3000
6 7 SRI HARSHAVARDHANAN R 6000

Employee DataFrame Information:
<class 'pandas.DataFrame'>
RangeIndex: 7 entries, 0 to 6
Data columns (total 3 columns): #
Column Non-Null Count Dtype ---
------ -------------- ----- 0 emp
id 7 non-null int64 1 name 7
non-null str  2 salary 7 non-null
int64 dtypes: int64(2), str(1)
memory usage: 410.0 bytes

Salary Column:
0 5000
1 6000
2 7000
3 5000
4 8000
5 3000
6 6000
Name: salary, dtype: int64

Mean Salary: 5714.285714285715
Median Salary: 6000.0
```

```
Mode Salary: [5000, 6000]
Salary Variance: 2571428.5714285714
Salary Standard Deviation: 1603.5674514745463

Statistical Summary:
 emp id salary
count 7.000000 7.000000
mean 4.000000 5714.285714
std 2.160247 1603.567451
min 1.000000 3000.000000
25% 2.500000 5000.000000
50% 4.000000 6000.000000
75% 5.500000 6500.000000
max 7.000000 8000.000000

Complete Summary:
 emp id name salary count 7.000000 7 7.000000 unique
NaN 6 NaN top NaN SRI HARSHAVARDHANAN R NaN freq NaN
2 NaN mean 4.000000 NaN 5714.285714 std 2.160247 NaN
1603.567451   min   1.000000   NaN   3000.000000   25%
2.500000    NaN    5000.000000    50%    4.000000    NaN
6000.000000   75%   5.500000   NaN   6500.000000   max
7.000000 NaN 8000.000000

Column Names:
Index(['emp id', 'name', 'salary'], dtype='str')

Employee Array:
[[1 'SREE VARSSINI K S' 5000]
 [2 'SREEMATHI B' 6000]
 [3 'SREYA G' 7000]
 [4 'SREYASKARI MULLAPUDI' 5000]
 [5 'SRI AKASH U G' 8000]
 [6 'SRI HARSHAVARDHANAN R' 3000]
 [7 'SRI HARSHAVARDHANAN R' 6000]]

Final Employee DataFrame:
 emp id name salary
0 1 SREE VARSSINI K S 5000
1 2 SREEMATHI B 6000
2 3 SREYA G 7000
3 4 SREYASKARI MULLAPUDI 5000
4 5 SRI AKASH U G 8000
5 6 SRI HARSHAVARDHANAN R 3000
6 7 SRI HARSHAVARDHANAN R 6000
```

In [ ]: